

|         |                                                                                                                                                                                                                                                                                                          |                    |     |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----|
| 3       | (a) charges on plates are equal and opposite<br>so no resultant charge<br>energy stored because there is charge separation                                                                                                                                                                               | M1<br>A1<br>B1     | [3] |
| (b) (i) | capacitance = $Q / V$<br>$= (18 \times 10^{-3}) / 10$<br>$= 1800 \mu\text{F}$                                                                                                                                                                                                                            | C1<br><br>A1       | [2] |
| (ii)    | use of area under graph <i>or</i> energy = $\frac{1}{2}CV^2$<br>energy = $2.5 \times 15.7 \times 10^{-3}$ <i>or</i> energy = $\frac{1}{2} \times 1800 \times 10^{-6} \times (10^2 - 7.5^2)$<br>$= 39 \text{ mJ}$                                                                                         | C1<br><br>A1       | [2] |
| (c)     | combined capacitance of Y & Z = $20 \mu\text{F}$ <i>or</i> total capacitance = $6.67 \mu\text{F}$<br>p.d. across capacitor X = $8 \text{ V}$ <i>or</i> p.d. across combination = $12 \text{ V}$<br>charge = $10 \times 10^{-6} \times 8$ <i>or</i> $6.67 \times 10^{-6} \times 12$<br>$= 80 \mu\text{C}$ | C1<br>C1<br><br>A1 | [3] |

|               |                                       |                 |              |
|---------------|---------------------------------------|-----------------|--------------|
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4. (c)  $\Delta U$  increases in internal energy.

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